

8. A shunt generator has the following open-circuit characteristic at 800 r.p.m.

Field amperes	:	0.5	1.0	1.5	2.0	2.5	3.0	3.5
E.M.F. Volt	:	54	107	152	185	210	230	245

Armature and shunt field resistances are respectively 0.1Ω and 80Ω . The terminal p.d. falls to 175 V when the armature current is 100 A. Find the O.C. volts and the volts lost due to (i) reduction in the field current (ii) armature resistance (iii) armature reaction. [220 V (i) 27 V (ii) 120 V (iii) 8 V]

9. The open-circuit characteristic of a shunt generator when driven at normal speed is as follows :

Field current	:	0.5	1.0	1.5	2.0	2.5	3.0	3.5 A
O.C. volts	:	54	107	152	185	210	230	240 V

The resistance of armature circuit is 0.1Ω . Due to armature reaction the effective field current is given by the relation $I_{sh}(\text{eff.}) = I_{sh} - 0.003 I_a$. Find the shunt field circuit resistance that will give a terminal voltage of 220 V with normal speed (a) on open circuit (b) at a load current of 100 A. Also find (c) number of series turns for level compounding at 220 V with 100 A armature current ; take number of shunt turns per pole as 1200 and (d) No. of series turns for over-compounding giving a terminal voltage of 220 V at no-load and 230 V with 100 A armature current. [(a) 80Ω (b) 66Ω (c) 6.8 Ω (d) 9.1 turns]

10. Find how many series turns per pole are needed on a 500-kW compound generator required to give 450 V on no-load and 500 V on full-load, the requisite number of ampere-turns per pole being 9,000 and 6,500 respectively. The shunt winding is designed to give 450 V at no-load when its temperature is 20°C . The final temperature is 60°C . Take $\alpha_0 = 1/234.5$ per $^\circ\text{C}$.

[2.76] (Electrical Technology, Allahabad Univ. 1977)

OBJECTIVE TESTS – 28

- The external characteristic of a shunt generator can be obtained directly from its — characteristic.
(a) internal (b) open-circuit
(c) load-saturation (d) performance
- Load saturation characteristic of a d.c. generator gives relation between
(a) V and I_a (b) E and I_a
(c) E_0 and I_f (d) V and I_f
- The slight curvature at the lower end of the O.C.C. of a self-excited d.c. generator is due to
(a) residual pole flux
(b) high armature speed
(c) magnetic inertia
(d) high field circuit resistance.
- For the voltage build-up of a self-excited d.c. generator, which of the following is not an essential condition ?
(a) There must be some residual flux
(b) Field winding mmf must aid the residual flux
(c) Total field circuit resistance must be less than the critical value
(d) Armature speed must be very high.
- The voltage build-up process of a d.c. generator is
(a) difficult (b) delayed
(c) cumulative (d) infinite
- Which of the following d.c. generator cannot build up on open-circuit ?
(a) shunt (b) series
(c) short shunt (d) long shunt
- If a self-excited d.c. generator after being installed, fails to build up on its first trial run, the first thing to do is to
(a) increase the field resistance
(b) check armature insulation
(c) reverse field connections
(d) increase the speed of prime mover.
- If residual magnetism of a shunt generator is destroyed accidentally, it may be restored by connecting its shunt field
(a) to earth
(b) to an a.c. source
(c) in reverse
(d) to a d.c. source.
- The three factors which cause decrease in the terminal voltage of a shunt generator are
(a) armature reactances
(b) armature resistance
(c) armature leakages
(d) armature reaction
(e) reduction in field current
- If field resistance of a d.c. shunt generator is increased beyond its critical value, the generator
(a) output voltage will exceed its name-plate rating
(b) will not build up

- (c) may burn out if loaded to its name-plate rating
(d) power output may exceed its name-plate rating
11. An ideal d.c. generator is one that has voltage regulation.
(a) low (b) zero
(c) positive (d) negative
12. The generator has poorest voltage regulation.
(a) series (b) shunt
(c) compound (d) high
13. The voltage regulation of an overcompound d.c. generator is always
- (a) positive
(b) negative
(c) zero
(d) high
14. Most commercial compound d.c. generator are normally supplied by the manufacturers as over compound machines because
(a) they are ideally suited for transmission of d.c. energy to remotely-located loads
(b) degree of compounding can be adjusted by using a diverter across series field
(c) they are more cost effective than shunt generators
(d) they have zero percent regulation.

ANSWERS

1. (b) 2. (d) 3. (c) 4. (d) 5. (c) 6. (b) 7. (c) 8. (d) 9. (b,d,e) 10. (b) 11. (b) 12. (a) 13. (b) 14. (b).

QUESTIONS AND ANSWERS ON D.C. GENERATORS

Q. 1. How may the number of parallel paths in an armature be increased ?

Ans. By increasing the number of magnetic poles.

Q. 2. How are brushes connected in a d.c. generator ?

Ans. Usually, all positive brushes are connected together and all the negative brushes together (Fig. 28.33).

Q. 3. What is meant by armature reaction ?

Ans. It is the effect of armature magnetic field on the distribution of flux under main poles of a generator.

The armature magnetic field has two effects :

- (i) It demagnetises or weakens the main flux and
- (ii) It cross-magnetises or distorts it.

Q. 4. What is the effect of this distortion on the operation of the machine ?

Ans. It acts as a magnetic drag on the armature which consequently requires more power to turn it.

Q. 5. How can field distortion be remedied ?

Ans. By using compensating windings which are embedded in the slots in the pole-shoe and are connected in series with the armature.

Q. 6. What is meant by normal neutral plane ?

Ans. It is a plane which passes through the axis of the armature perpendicular to the magnetic field of the generator when there is no flow of current through the armature.

Q. 7. What is the importance of this plane in the working of the machine ?

Ans. It gives the position where brushes would be placed to prevent sparking during the operation of the generator where the main pole field not distorted by armature field and were there no self-induction in the coils.

Q. 8. How do you differentiate between normal neutral plane ?

Ans. The *NNP* is the position of zero induction and hence minimum sparking assuming no field distortion i.e. on no-load. It is perpendicular to magnetic axis. *NA* is the position of zero induction and hence minimum sparking with distorted field i.e. **when generator is on load.**

Q. 9. How do you define 'commutating plane' ?

Ans. It is the plane which passes through the axis of the armature and through centre of contact of the brushes as shown in Fig. 28.34.

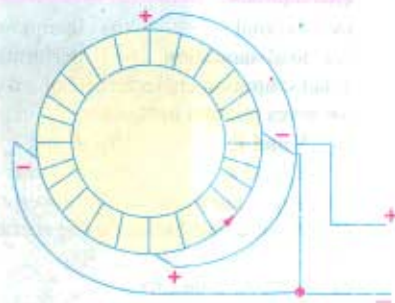


Fig. 28.33